

Astrodynamics I: Midterm Exam

April 28, 2012 9:00-10:40AM (100 minutes)

Use the following constants and formulas if necessary.

- $\vec{e} = \frac{(v^2 - \frac{\mu}{r})\vec{r} - (\vec{r} \cdot \vec{v})\vec{v}}{\mu}$, $e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$

- $\mu_E = 3.986 \times 10^5 km^3/sec^2$, $R_E = 6,378 km$

1. (30) An Earth-orbiting artificial satellite in elliptical orbit is observed to have an altitude at perigee of 220 km and an altitude at apogee of 360 km.
 - a. (4) What is the orbital specific angular momentum?
 - b. (4) What is the orbital specific energy?
 - c. (4) What is the orbital eccentricity?
 - d. (4) What are the orbital velocities at perigee and at apogee?
 - e. (4) What is the orbital period?
 - f. (10) At a true anomaly of 45 degrees as measured from perigee, determine the orbital altitude, the magnitude of the orbital velocity vector, and the flight path angle.
2. (30) Describe how to obtain $(\vec{r}, \vec{v})$ from the classical orbital elements $(a, e, i, \Omega, \omega, \nu)$.
3. (30) Given $\vec{r} = -\hat{i} - \hat{j} - \hat{k}$ and $\vec{v} = \frac{1}{3}(\hat{i} - \hat{j} + \hat{k})$, determine the orbital elements and sketch the orbit. Use $\mu=1$.
4. (10) Discuss the Integrals of Motion for the 2 body problem.